Instance	1	[illegible]	[illegible]	[illegible]	[illegible]	0	0	0	1	05

Department of Computer Science and Engineering

Faculty of Engineering & Technology, Gurukula Kangri Deemed to be University, Haridwar

First Sessional Exam

Subject: Machine Learning

Paper Code: BCE-T517

Time: 1 Hr MM: 20

Note: Answer any three questions from Section A (each question carries 4 marks).
Answer any one question from Section B (each question carries 8 marks).

Q.No.	Section A (Questions of 4 marks each)	Keyword	CO	BT
Q5.	What is the primary difference between Artificial Intelligence (AI) and Machine Learning (ML)?	Remembering	(CO1)	BT1
Q6.	Explain what is meant by "Batch learning" in Machine Learning.	Applying	(CO4)	BT3
Q7.	Describe one application of Machine Learning in the field of healthcare.	Understanding	(CO2)	BT2
Q8.	Compare and contrast Structured Data and Unstructured Data in Machine Learning.	Analyzing	(CO6)	BT4
	Section B (Questions of 8 marks each)			
Q3.	Discuss the types of problems in Machine Learning, such as Regression and Classification, and give examples of real-world scenarios where each type is applied.	Applying	(CO3)	BT3
Q4.	What are the limitations of Machine Learning algorithms? Discuss the importance of understanding these limitations when implementing ML in an organization.	Evaluating	(CO4)	BT5

Department of Computer Science and Engineering
Faculty of Engineering & Technology, Gurukula Kangri Deemed to be University, Haridwar
Second Sessional Exam
Subject: Machine Learning
Paper Code: BCE-P517

Time: 1 Hr **MM: 20**

Note: Answer any two questions from Section A (each question carries 5 marks).
Answer any one question from Section B (each question carries 10 marks).

Q.No.	Section A (Questions of 5 marks each)	Keyword	CO	BT
Q1.	Actual values: [3,4,5,6] Predicted values: [2.5,4.2,4.8,6.3]. Calculate Error Metrics: i) MAE ii) SSE iii)MSE iv) RMSE v)MAPE vi) R-Squared vii)Adjusted R-Square Error	Applying	(CO3)	BT3
Q2.	Input:{Cat, Dog, Birds}, logits:Z_{cat}=2.0 Z_{dog}= 1.0 Z_{Birds} =0.1,Find the Probabilities of the 3 classes Softmax Function.	Applying	(CO4)	BT3
Q3.	We are doing a multi label classification with following cat:1.5 dog:0.5 Birds:2.0, logits from this model what classes does the input level correctly if we use a threshold probability of 70% (Use Sigmoid function to calculate probabilities)	Applying	(CO4)	BT3
	Section B (Questions of 10 marks each)			
Q4	X1=[1,2,3,4] X2=[1,3,4,6] Y=[1,7,9,13] x1 an x2 are Input → y as actual o/p Y=βo+X1β1+X2β2 Find the optimal value of β using least Square method and Y	Applying	(CO3)	BT3
Q5	See table below. Find TPR FPR ROC and AUC	Applying	(CO3)	BT3

Instance	1	2	3	4	5	6	7	8	9	10		
Actual	1	1	1	0	1	0	0	0	1	0		
Probability	.9	.8	.7	.6	.5	.4	.3	.2	.1	.05		
Threshold	1	.95	.85	.75	.65	.55	.45	.35	.25	1.5	.5	0

Find TPR FPR ROC and AUC

Find TPR, FPR, ROC and AUC

Department of Computer Science and Engineering
Faculty of Engineering & Technology, Gurukula Kangri Deemed to be University, Haridwar
Third Sessional Exam
Subject: Machine Learning
Paper Code: BCE-P517

Time: 1 Hr **MM: 20**

Note: Answer any two questions from Section A (each question carries 5 marks).
Answer any one question from Section B (each question carries 10 marks).

Q.No.	Section A (Questions of 5 marks each)
Q1.	A bag contains 10md balls and 15 blue balls. Two balls are drawn randomly without replacerat. Given that the first ball drawn is red, the probability (rounded off to decimal places) that both balls drawn are red is
Q2.	What is the role of the Numpy library in machine learning, and how does it differ from Pandas in terms of functionality and use cases?
Q3.	A matrix A and one of its eigenvectors are given. Find the eigenvalue of A for the given A={{9 8}{-6 -5}} $\vec{x}$ ={{−4 } {3}}
	Section B (Questions of 10 marks each)
Q1.	Let A denote the event that the midtown temperature in Los Angeles is 70°F, and let B denote the event that the midtown temperature in New York is 70°F. Also, let C denote the event that the maximum of the midtown temperatures in New York and in Los Angeles is 70°F. If P(A) = 3 P(B)= 4, and P(C) = 0.2 find the probability that the minimum of the two midtown temperatures is 70°F
Q2.	What are the advantages of using cloud-based platforms like Google Collab for training machine learning models, especially when working with large datasets?